

OSBAMS Rev.2 controller PCB - schematic package

Source: Hardware/Schematic/OSBAMS PCB.kicad_sch (KiCad format 20260306) | generated 2026-10-01 by tools/mfg

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Contents

1. Complete schematic (single sheet). 2-5. The four schematic sections. 6. Power tree. 7. Section coverage vs the requested list. 8. Netlist and component list.

What this board is

A through-hole carrier board: 12 V input with reverse-polarity diode and TVS, a low-side MOSFET driver for the contactor coil with E-stop loop and flyback diode, a TC74 temperature sensor, a header for an external INA228 breakout, a 2x4 header to an external NUCLEO-L476RG and a 4-pin UART header. Net connectivity of the schematic and PCB was compared by script: identical (0 differences).

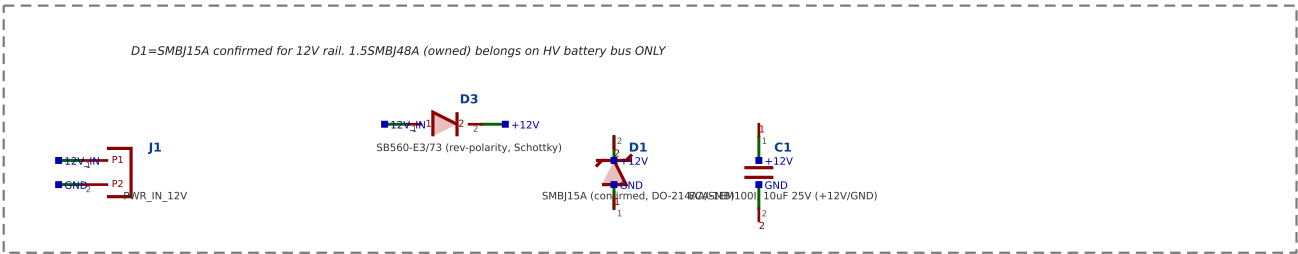
Schematic observations

- Diode symbol (OSBAMS:D_H, OSBAMS:D_TVS): pin 1 is the ANODE (triangle base) and pin 2 the CATHODE. KiCad's D_DO-201AD and D_SMB footprints use pad 1 = CATHODE. The PCB therefore connects each diode's cathode to the net the schematic gives its anode. See PCB review, finding R1.
- No title block (sheet title, revision, date, company) is present in the schematic.
- Value text of D1 and C1 overlap on the sheet (cosmetic).
- J5 is annotated 'keyed' but uses a plain header footprint.
- E-stop appears only as a loop (J3) in series with the coil supply; no E-stop status signal reaches the MCU header.
- No ADC measurement circuit, no USB, no 5 V rail and no test points are drawn.

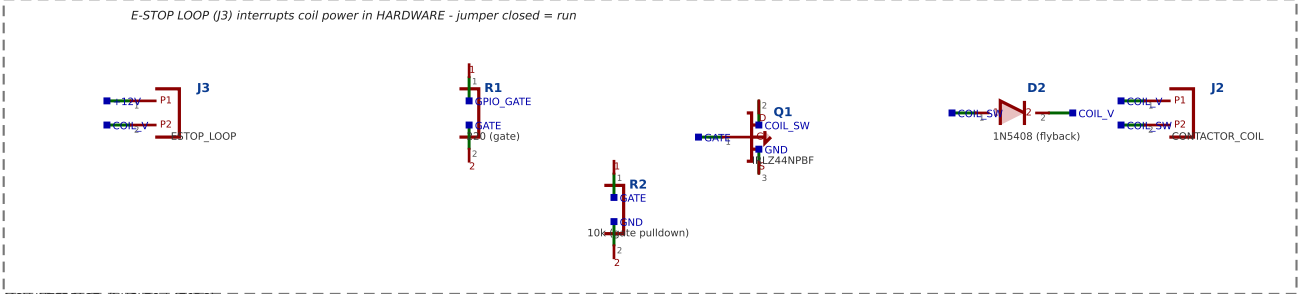
OSBAMS Rev.2 controller - Complete schematic (single sheet)

All sections. Label-based connectivity: nets of the same name are connected.

12V POWER INPUT + PROTECTION



SAFETY / CONTACTOR DRIVER

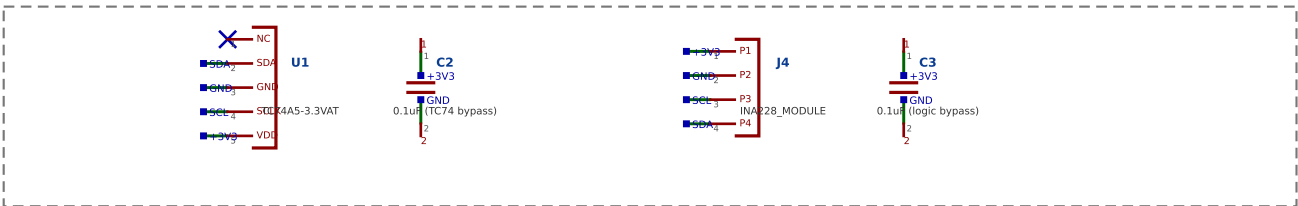


MCU INTERFACE (NUCLEO-L476RG)



J5 = NUCLEO_INTERFACE: 2x4 keyed connector to external Nucleo-L476RG daughterboard. Pin8=SPARE_GPIO reserved, unconnected on this board.

SENSORS (I2C BUS)

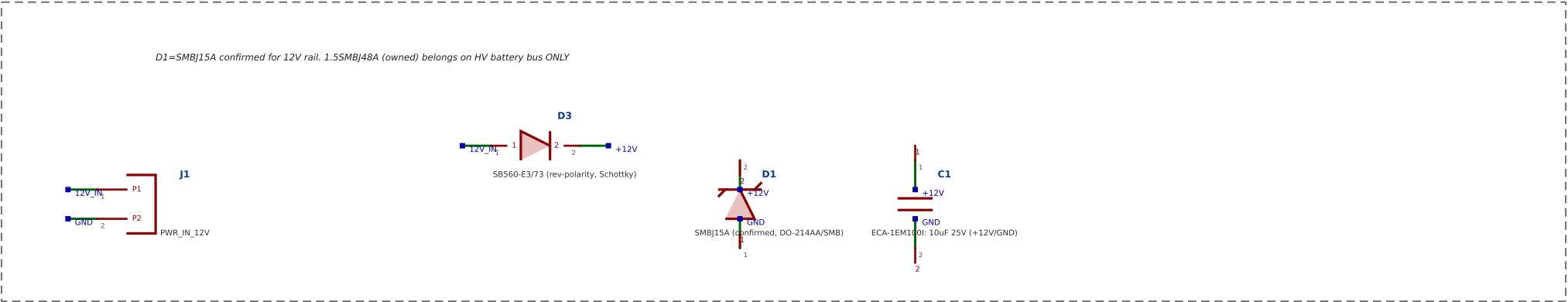


+3V3 comes from Nucleo 3.3V rail. TC74 addr=0x4D, INA228 addr=0x40 - no conflict

OSBAMS Rev.2 controller - Section 1 - 12 V power input and protection (J1, D3, D1, C1)

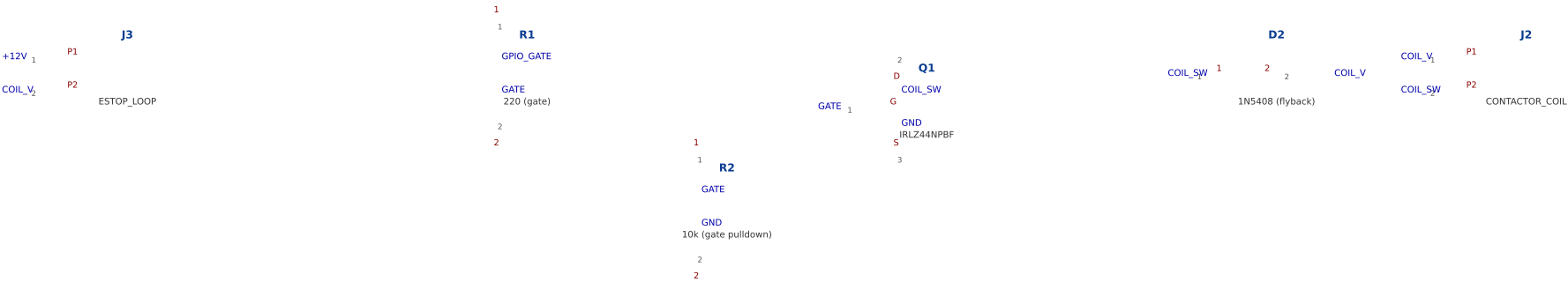
Reverse-polarity Schottky D3, TVS D1, bulk capacitor C1. NOTE diode pin 1 = ANODE in this symbol library.

12V POWER INPUT + PROTECTION



SAFETY / CONTACTOR DRIVER

E-STOP LOOP (J3) interrupts coil power in HARDWARE - jumper closed = run

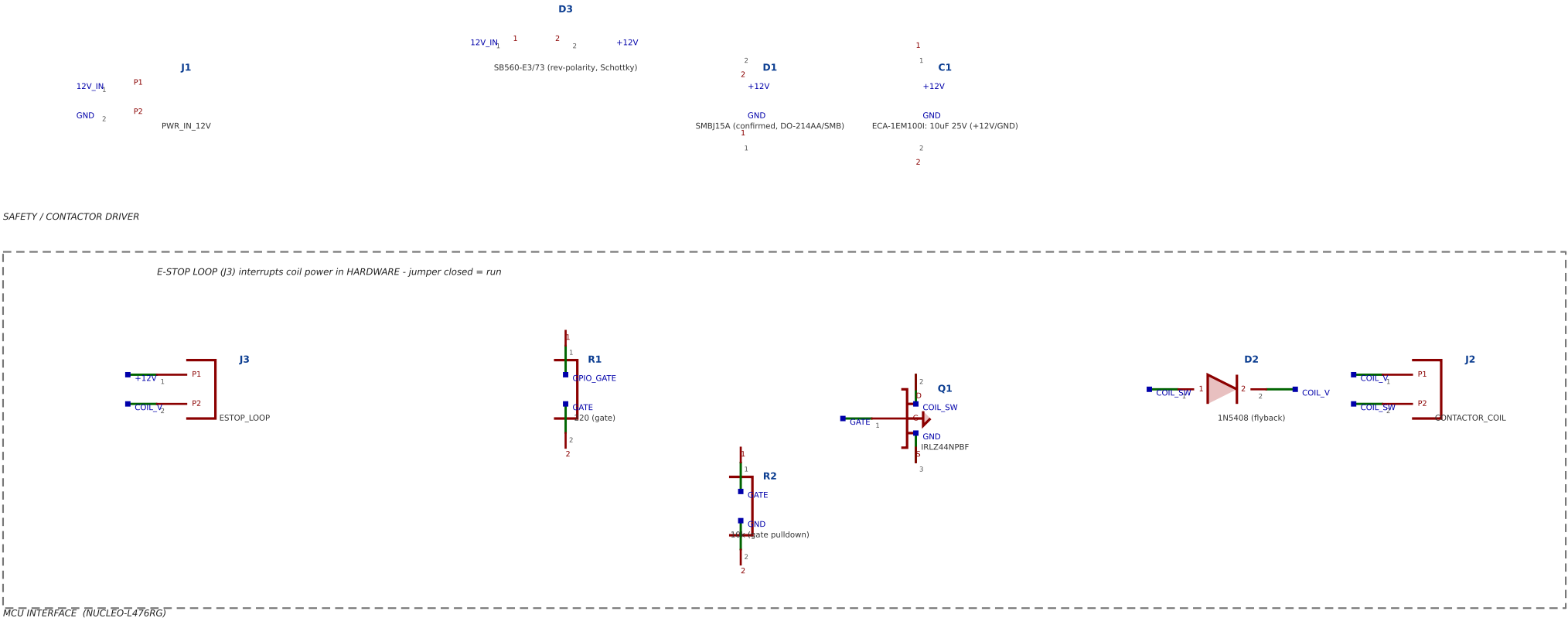


OSBAMS Rev.2 controller - Section 2 - Relay (contactor-coil) driver and E-stop loop (J3, R1, R2, Q1, D2, J2)

12V POWER INPUT + PROTECTION

Low-side MOSFET switch; E-stop loop J3 in series with the coil supply (hardware interruption).

D1=SMBJ15A confirmed for 12V rail. 1.5SMBJ48A (owned) belongs on HV battery bus ONLY



MCU INTERFACE (NUCLEO-L476RG)

NUCLEO_INTERFACE

+3V3₁ P1
GND₂ P2
SDA₃ P3
SCL₄ P4
GPIO_GATE P5
UART_TX P6
UART_RX P7
SPARE_GPIO P8

J5

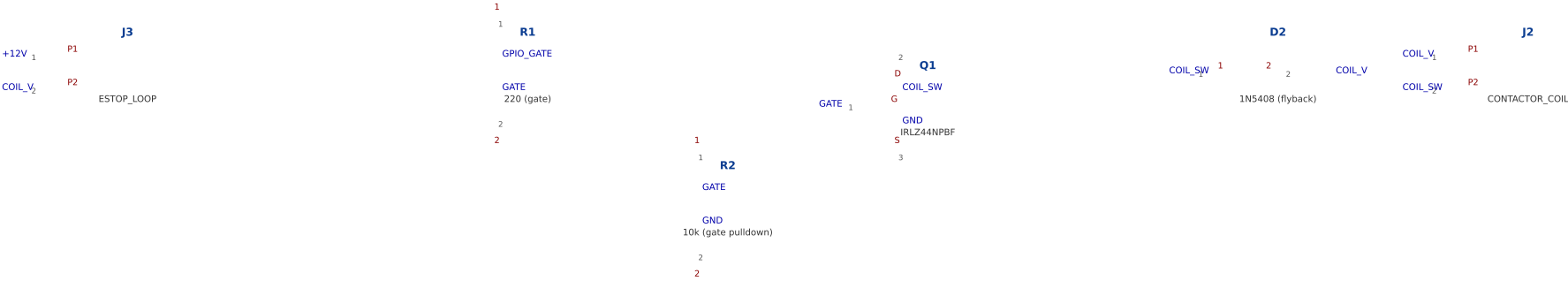
+3V3₁ P1
GND₂ P2
UART_TX P3
UART_RX P4

J6
UART_HDR

OSBAMS Rev.2 controller - Section 3 - MCU interface and UART/Pi header (J5, J6)

E-STOP LOOP (J3) interrupts coil power in HARDWARE - Jumper closed = run

STM32L476RG lives on the external NUCLEO-L476RG; only this header is on the PCB. No USB, no ADC channel.



MCU INTERFACE (NUCLEO-L476RG)



SENSORS (I2C BUS)



+3V3 comes from Nucleo 3.3V rail. TC74 addr=0x4D, INA228 addr=0x40 - no conflict

OSBAMS Rev.2 controller - Section 4 - Sensors: temperature (U1) and INA228 module header (J4)

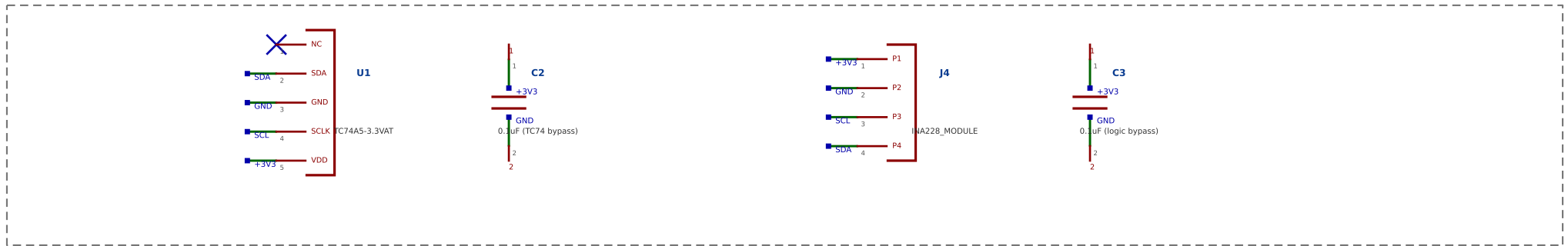
I2C bus: TC74 address 0x4D, INA228 address 0x40.

NUCLEO_INTERFACE

+3V3 1 P1
GND 2 P2
SDA 3 P3
SCL 4 P4
GPIO_GATE P5
UART_TX P6
UART_RX P7
SPARE_GPIO P8
J5

+3V3 1 P1
GND 2 P2
UART_TX P3
UART_RX P4
J6
UART_HDR

J5 = NUCLEO_INTERFACE: 2x4 keyed connector to external Nucleo-L476RG daughterboard. Pin8=SPARE_GPIO reserved, unconnected on this board.
SENSORS (I2C BUS)

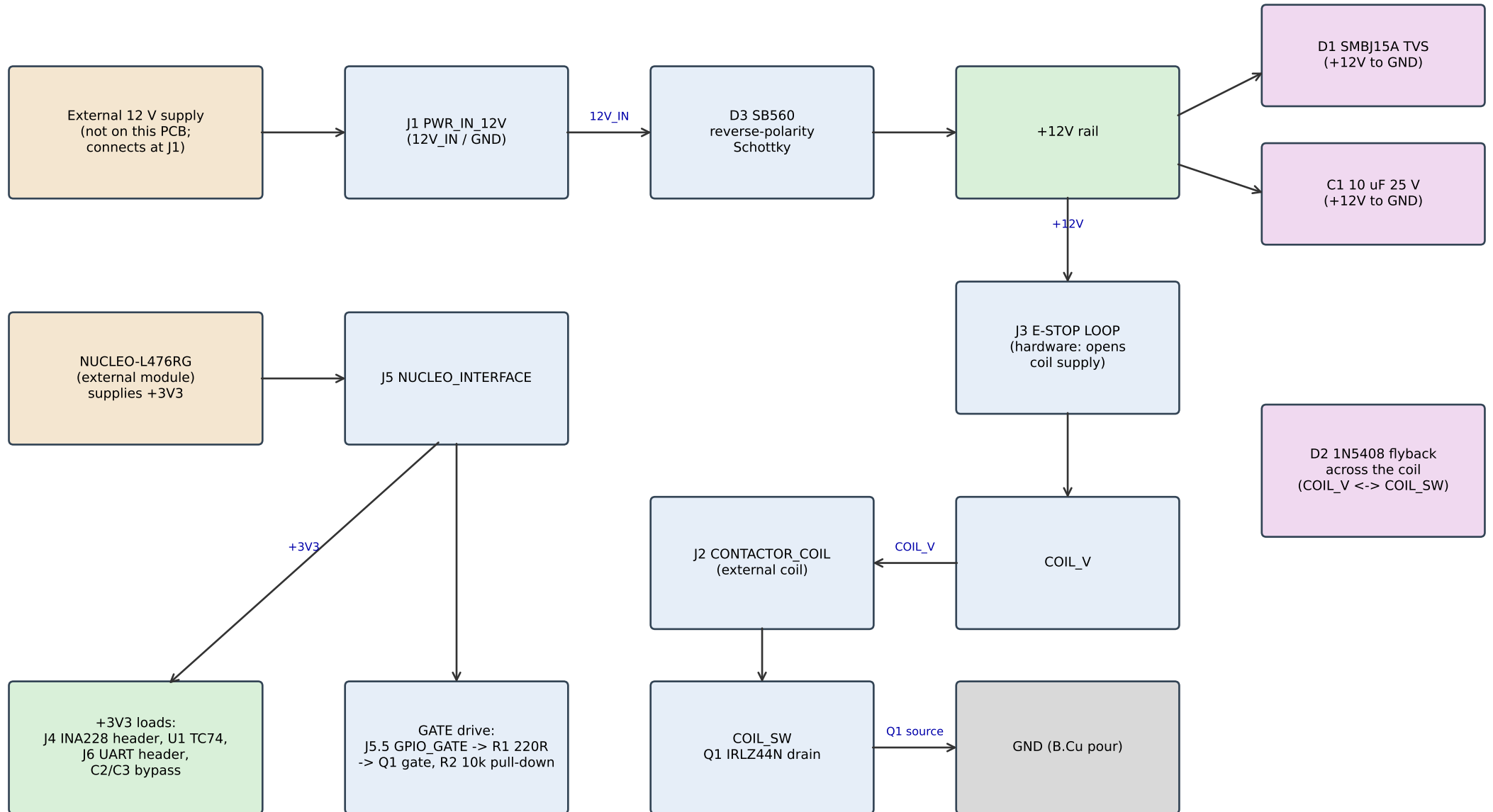


+3V3 comes from Nucleo 3.3V rail. TC74 addr=0x4D, INA228 addr=0x40 - no conflict

OSBAMS Rev.2 controller PCB - power tree

Derived from the netlist. There is NO 5 V rail and NO regulator on this PCB; 3.3 V comes from the Nucleo module.

No ADC divider, no E-stop sense line, no relay-feedback input, no 5 V rail on this PCB.



Section coverage against the requested schematic contents

Requested section	Present on this PCB?	Where	Comment
Power tree	Yes	Section 1 / power-tree page	12 V in -> D3 -> +12V; no 5 V; 3V3 from Nucleo
STM32 section	Interface only	J5	STM32L476RG is on the external Nucleo module
INA228 section	Interface only	J4	INA228 is an external breakout; header carries 3V3/GND/SCL/SDA only. Shunt/VBUS wiring is off-board.
ADC measurement section	NO	-	No ADC divider or ADC input on this PCB
Temperature interface	Yes	U1, C2	TC74A5-3.3VAT, I2C 0x4D
UART / USB / Pi interface	UART header only	J6 (+ J5.6/J5.7)	3.3 V UART; no USB connector
Relay driver	Yes	Q1, R1, R2, D2, J2	Low-side MOSFET, gate pull-down, flyback diode
E-stop monitoring	Loop only	J3	Opens coil supply in hardware; NO status line to the MCU
Protection circuits	Yes	D3, D1, D2, C1	Reverse-polarity Schottky, TVS, flyback, bulk cap (polarity issue R1)

Netlist (from the schematic)

Net	Pins
+12V	C1.1, D1.2, D3.2, J3.1
+3V3	C2.1, C3.1, J4.1, J5.1, J6.1, U1.5
12V_IN	D3.1, J1.1
	U1.1
COIL_SW	D2.1, J2.2, Q1.2
COIL_V	D2.2, J2.1, J3.2
GATE	Q1.1, R1.2, R2.1
GND	C1.2, C2.2, C3.2, D1.1, J1.2, J4.2, J5.2, J6.2, Q1.3, R2.2, U1.3
GPIO_GATE	J5.5, R1.1
SCL	J4.3, J5.4, U1.4
SDA	J4.4, J5.3, U1.2
SPARE_GPIO	J5.8
UART_RX	J5.7, J6.4
UART_TX	J5.6, J6.3

Components (from the schematic)

Ref	Value (as in schematic)	Footprint
C1	ECA-1EM100I: 10uF 25V (+12V/GND)	Capacitor_THT:CP_Radial_D5.0mm_P2.00mm
C2	0.1uF (TC74 bypass)	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
C3	0.1uF (logic bypass)	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm
D1	SMBJ15A (confirmed, DO-214AA/SMB)	Diode_SMD:D_SMB
D2	1N5408 (flyback)	Diode_THT:D_DO-201AD_P15.24mm_Horizontal
D3	SB560-E3/73 (rev-polarity, Schottky)	Diode_THT:D_DO-201AD_P15.24mm_Horizontal
J1	PWR_IN_12V	TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
J2	CONTACTOR_COIL	TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
J3	ESTOP_LOOP	TerminalBlock:TerminalBlock_MaiXu_MX126-5.0-02P_1x02_P5.00mm
J4	INA228_MODULE	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical
J5	NUCLEO_INTERFACE	Connector_PinHeader_2.54mm:PinHeader_2x04_P2.54mm_Vertical
J6	UART_HDR	Connector:FanPinHeader_1x04_P2.54mm_Vertical
Q1	IRLZ44NPBF	Package_TO_SOT_THT:TO-220-3_Vertical
R1	220 (gate)	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal

Ref	Value (as in schematic)	Footprint
R2	10k (gate pulldown)	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
U1	TC74A5-3.3VAT	Package_TO_SOT_THT:TO-220-5_Vertical